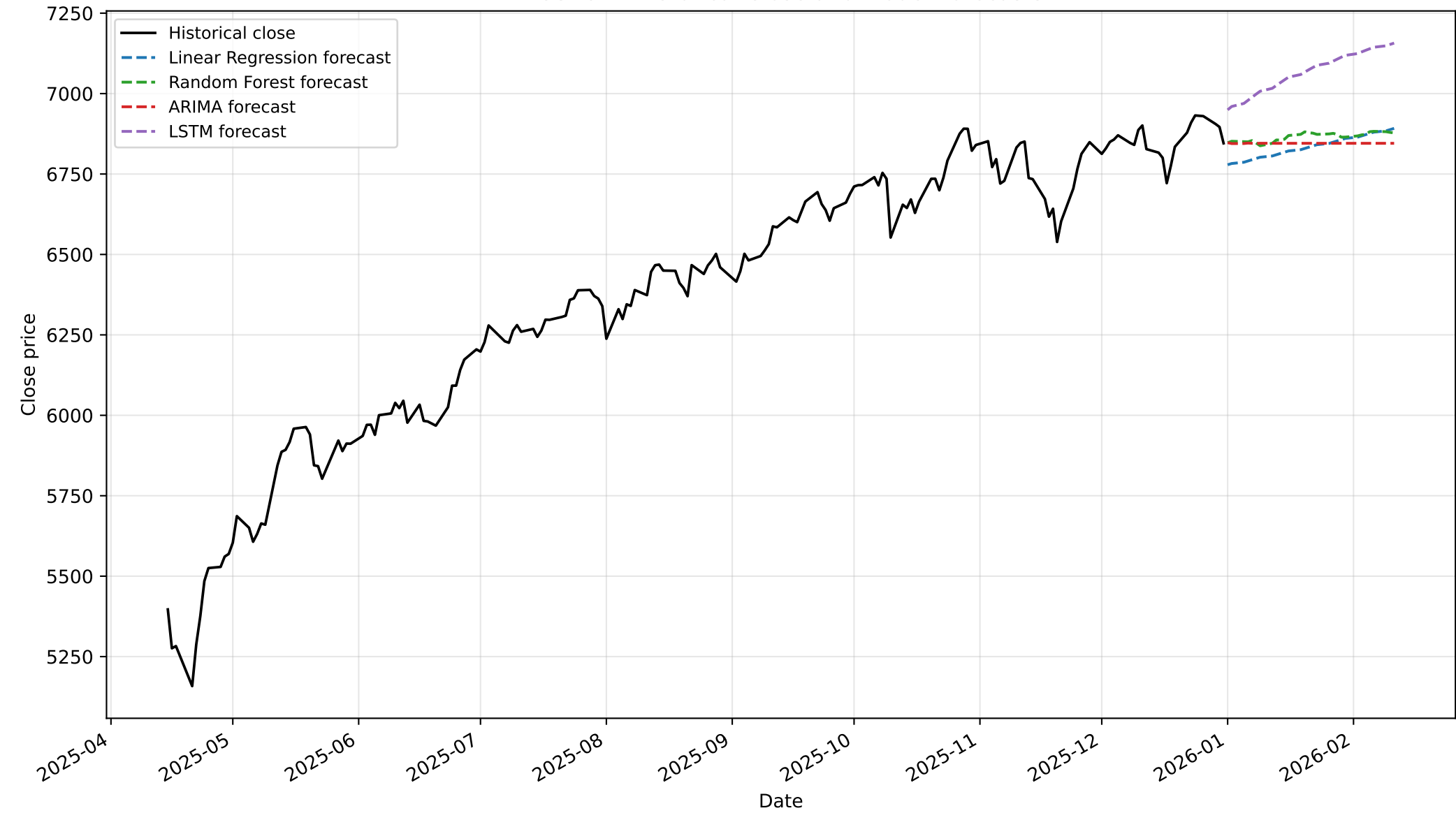


PDA3 Batch Analysis Report
Symbol: ^GSPC
Source file: ^GSPC260101.csv
Generated: 2026-09-26 03:05 UTC
Observations used: 752

Latest indicators
Last Close: 6845.5
Change 30D: 2.59
Sma 20: 6856.23
Sma 50: 6802.55
Volatility: 15.07

Estimates are educational model outputs, not investment advice. Historical accuracy does not guarantee future performance.

^GSPC — Historical Close and Model Forecasts



Model validation metrics (walk-forward holdout)

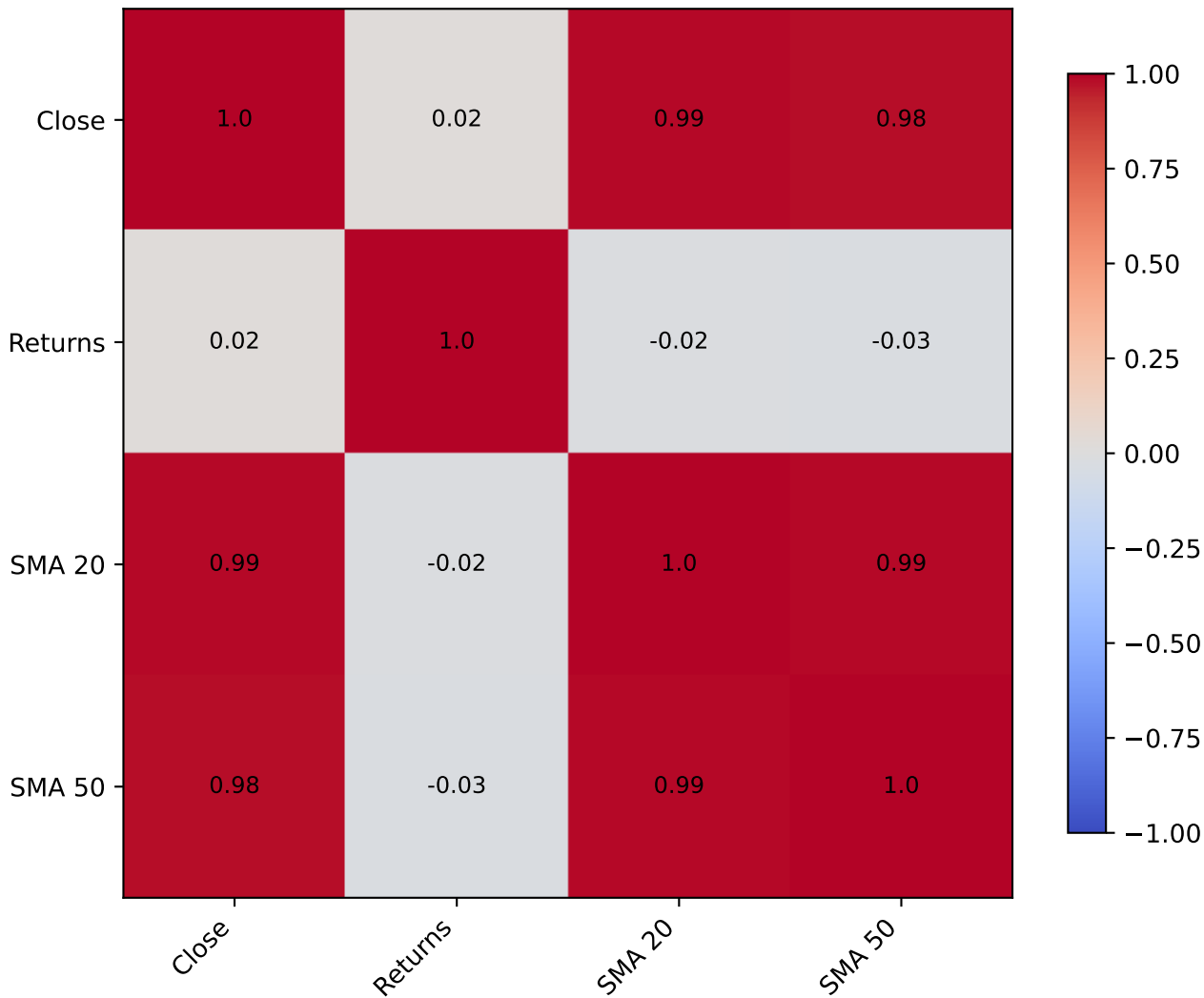
Model	MAE	RMSE	MAPE (%)	R ²	Notes
Linear Regression	121.857	133.697	1.78	-0.8341	
Random Forest	66.673	89.457	0.99	0.1789	
ARIMA	155.596	167.91	2.27	-1.8929	
LSTM	252.297	263.478	3.71	-6.1231	

R² below 0 means a model performed worse than simply predicting the average price over the validation window — common when a stock reverses or plateaus after a strong trend.

Forecast table (next trading days)

Date	Linear Regression	Random Forest	ARIMA	LSTM
2026-01-01	6778.78	6847.26	6847.47	6949.36
2026-01-02	6782.68	6851.88	6844.64	6959.97
2026-01-05	6786.58	6850.61	6844.84	6969.4
2026-01-06	6790.49	6850.08	6846.15	6978.96
2026-01-07	6794.39	6854.04	6845.51	6988.59
2026-01-08	6798.29	6846.77	6845.13	6998.12
2026-01-09	6802.19	6837.74	6845.6	7007.36
2026-01-12	6806.09	6845.62	6845.6	7016.36
2026-01-13	6809.99	6855.91	6845.37	7025.36
2026-01-14	6813.89	6855.37	6845.46	7034.13
2026-01-15	6817.79	6858.24	6845.54	7042.69
2026-01-16	6821.7	6869.74	6845.46	7051.2
2026-01-19	6825.6	6873.23	6845.45	7059.29
2026-01-20	6829.5	6881.26	6845.49	7066.95
2026-01-21	6833.4	6878.17	6845.48	7074.26
2026-01-22	6837.3	6877.09	6845.47	7081.26
2026-01-23	6841.2	6873.2	6845.48	7087.99
2026-01-26	6845.1	6874.67	6845.48	7094.52
2026-01-27	6849.01	6876.43	6845.47	7100.92
2026-01-28	6852.91	6872.78	6845.48	7107.12
2026-01-29	6856.81	6864.48	6845.48	7113.26
2026-01-30	6860.71	6864.82	6845.48	7118.89
2026-02-02	6864.61	6868.45	6845.48	7124.32
2026-02-03	6868.51	6871.69	6845.48	7129.54
2026-02-04	6872.41	6873.35	6845.48	7134.57
2026-02-05	6876.31	6881.6	6845.48	7139.4
2026-02-06	6880.22	6882.69	6845.48	7144.05
2026-02-09	6884.12	6881.59	6845.48	7148.52
2026-02-10	6888.02	6879.38	6845.48	7152.81
2026-02-11	6891.92	6877.84	6845.48	7156.94

Correlation matrix

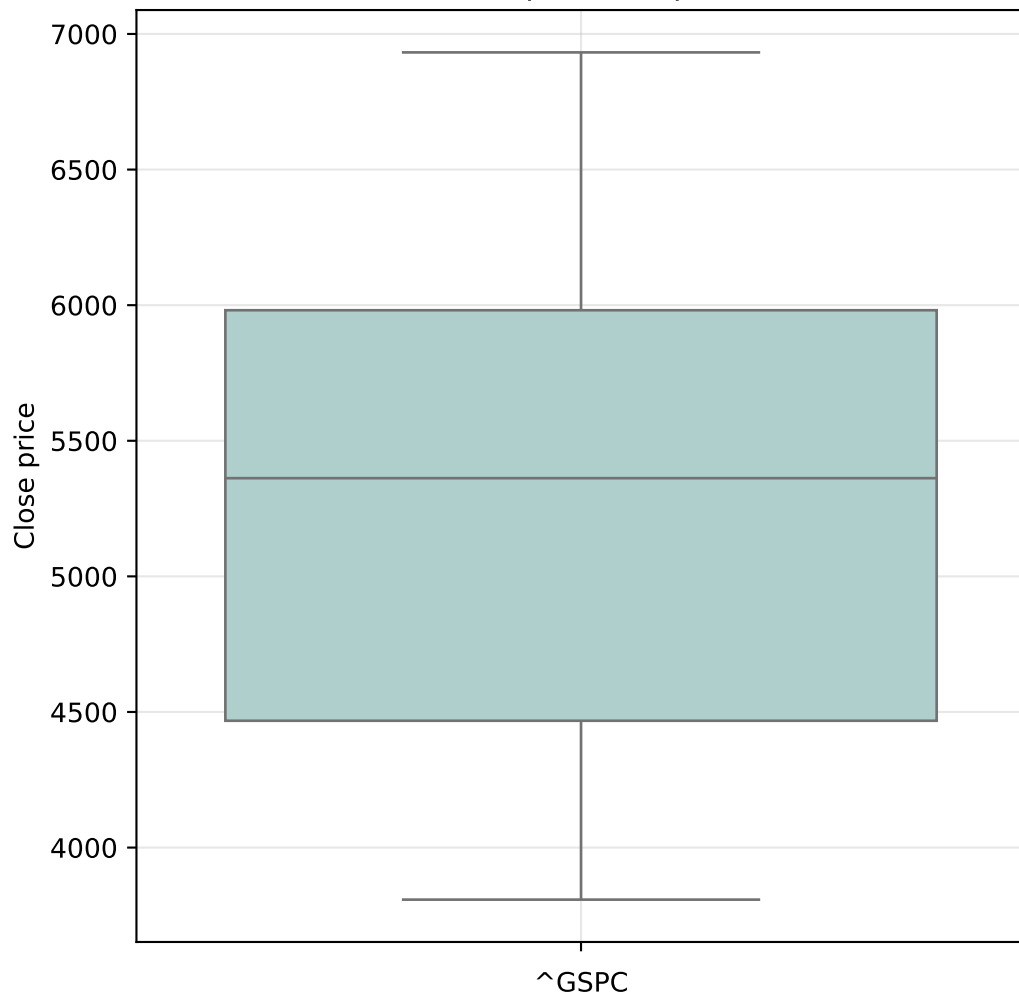


^GSPC — Source data statistics (close price)

Statistic	Value
Count	752
Mean	5309.926
Median	5362.075
Std Dev	873.397
Min	3808.1
Max	6932.05
Range	3123.95
Q1 (25%)	4467.642
Q3 (75%)	5981.333
IQR	1513.69
Skewness	0.053
Kurtosis	-1.155
Coeff. of Variation (%)	16.448

$\hat{GSPC}$ — Source data range and distribution

Close price boxplot



Close price distribution (violin plot)

